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324751 (24)

BE (7th Semester)

Examination, April-May, 2014

Branch : Electrical, EEE

POWER APPARATUS SYSTEM

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) All questions carry equal marks.

(ii) Part (a) of each question is compulsory.

(iii) Attempt any two parts from (b), (c) & (d).

UNIT - I

Q. 1. (a) What is insulator flashover. 2

(b) In a 33 KV overhead line, there are three units in the string of insulator. If the capacitance

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between insulator pin & earth is 11% of self capacitance of each insulator, find :

(i) distribution of voltage over 3 insulators

(ii) string efficiency 7

(c) An overhead transmission line conductor having a parabolic configuration weighs 1.925 kg per metre of length. The area of cross-section of conductor is 2.2 cm^2 & the ultimate strength is 8000 kg/cm^2. The supports are 600 m apart having 15 m difference of levels. Calculate the sag from the taller of two

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supports which must be allowed so that FOS shall be 5. Assume that ice-load is 1 kg/metre run and there is no wind pressure. 7

(d) Explain various means by which voltage distribution across an insulator string can be improved. 7

UNIT - II

Q. 2. (a) State the function of isolator. 2

(b) Give the classification of substations according to service & constructional features. 7

(c) Discuss various steps involved in design of earthing grid. 7

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(d) Draw & explain the single line diagram, showing the location of substation equipment for main transfer bus-bar scheme. 7

UNIT - III

Q. 3. (a) Why earthing is needed for equipments ? 2

(b) Explain solidly grounded system with phasor diagram. 7

(c) Discuss the earthing transformer. 7

(d) A 132 KV, 3ϕ, 50 cycles, 50 km overhead line has capacitance to earth for each line of 0.015 μF/m. Determine L & KVA rating of arc supression coil suitable for this system 7

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UNIT - IV

Q. 4. (a) Define surge absorber. 2

(b) Explain the mechanism of lightning discharge. 7

(c) Explain why insulation coordination is required in a large power system. 7

(d) Explain the protective characteristics of lightning arrester. 7

UNIT - V

Q. 5. (a) Define probability density function. 2

(b) Describe the bath tub curve with help of neat diagram & for which region are the reliability calculations valid & why ? 7

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(c) Using approximate method, find various reliability indices for a system having n components in series. 7

(d) Explain two state fluctuating environment. State its utility. 7

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